FILMORA

TESTING & QUALITY ASSURANCE SPECIFICATION

Project Name: FILMORA
Project Type: Free Online Movie Streaming Platform
Document Number: FILMORA-QA-006
Document Version: 1.0
Status: Proposed Design
Testing Approach: Functional, Integration, Security and Performance Testing
Frontend: HTML5, CSS3 and JavaScript
Backend: Node.js + Express.js
Database: PostgreSQL

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1. INTRODUCTION

The purpose of this document is to establish the testing and quality assurance requirements for FILMORA.

Testing will ensure that all platform components operate correctly, securely and reliably.

The testing process will cover the frontend, backend, database, authentication system, administrative dashboard and video streaming infrastructure.

2. QUALITY ASSURANCE OBJECTIVES

The main objectives are:

1. Identify software defects before production.
2. Verify that all features work correctly.
3. Ensure secure user authentication.
4. Verify database consistency.
5. Ensure reliable movie playback.
6. Verify administrator permissions.
7. Identify performance problems.
8. Ensure mobile responsiveness.
9. Maintain a consistent user experience.
10. Reduce production failures.

3. TESTING STRATEGY

FILMORA will use multiple testing levels.

Testing Type	Purpose
Unit Testing	Test individual functions
Integration Testing	Test connected modules
Functional Testing	Verify platform features
API Testing	Verify backend endpoints
Security Testing	Identify vulnerabilities
Performance Testing	Measure system performance
UI Testing	Verify interface behavior
Acceptance Testing	Confirm business requirements

Testing must be performed throughout development.

4. TESTING ENVIRONMENT

The project will use separate environments.

4.1 Development Environment

Used by developers during implementation.

4.2 Testing Environment

Used to verify features before release.

4.3 Production Environment

Used by real FILMORA users.

The testing environment must not contain real users' sensitive personal information.

5. UNIT TESTING

Unit testing will verify individual functions independently.

Examples:

- Password validation.
- Email validation.
- Movie data validation.

- Search filtering.
- Rating calculations.
- Viewing progress calculations.
- Permission checking.
- Date formatting.

Each function should produce the expected result for valid and invalid inputs.

6. INTEGRATION TESTING

Integration testing will verify communication between different system components.

Required tests:

- Frontend to backend communication.
- Backend to PostgreSQL communication.
- Authentication to session management.
- Movie management to database.
- Media management to storage.
- Streaming authorization to video delivery.
- Watch history to playback progress.

7. FUNCTIONAL TESTING

Functional testing will verify that platform features operate according to their specifications.

Main functional areas:

- Account registration.
- Login and logout.
- Movie discovery.
- Movie searching.
- Movie details.
- Video playback.
- Watchlists.
- Favorites.
- Reviews.
- User profiles.

- Administration.

8. AUTHENTICATION TESTING

The authentication system must be tested thoroughly.

8.1 Registration tests

- Register with valid information.
- Register with an existing email.
- Register with an existing username.
- Submit invalid email information.
- Submit weak passwords.
- Submit mismatched passwords.
- Attempt registration with missing fields.

8.2 Login tests

- Login with correct credentials.
- Login with incorrect credentials.
- Login with an unverified account.
- Login with a suspended account.
- Attempt repeated failed logins.
- Verify session creation.

8.3 Password recovery tests

- Request password recovery.
- Submit an unknown email.
- Use a valid reset token.
- Use an expired reset token.
- Attempt password changes.

9. USER MANAGEMENT TESTING

Verify that users can:

- View their profiles.
- Update personal information.
- Change passwords.

- Update profile pictures.
- Manage sessions.
- Delete their accounts.
- View their personal activity.

Also verify that one user cannot modify another user's private information without authorization.

10. MOVIE MANAGEMENT TESTING

Verify that authorized administrators can:

- Add movies.
- Edit movie information.
- Archive movies.
- Publish movies.
- Unpublish movies.
- Assign genres.
- Add cast information.
- Upload posters.
- Upload backdrops.

Verify that unauthorized users cannot perform administrative operations.

11. VIDEO STREAMING TESTING

Video streaming is a critical part of FILMORA.

Testing must verify:

- Authorized playback.
- Unauthorized playback rejection.
- Video loading.
- Play and pause.
- Fullscreen playback.
- Volume control.
- Seeking.
- Subtitle selection.
- Audio selection.

- Adaptive quality selection.
- Resume watching.
- Playback progress synchronization.
- Streaming interruption recovery.

Test different video qualities and network conditions.

Only properly licensed test videos should be used.

12. SEARCH AND DISCOVERY TESTING

Verify that users can:

- Search by movie title.
- Search by genre.
- Filter by release year.
- Filter by language.
- Search by cast member.
- Sort search results.
- Navigate paginated results.

Verify that empty searches and unavailable results are handled correctly.

13. WATCHLIST AND FAVORITES TESTING

Tests must verify:

- Adding movies to watchlists.
- Removing movies.
- Adding favorites.
- Removing favorites.
- Retrieving saved movies.
- Preventing duplicate entries.
- Maintaining user-specific collections.

14. WATCH HISTORY TESTING

Verify:

- Recording viewing activity.
- Saving playback progress.

- Updating progress.
- Resuming a movie.
- Marking completed movies.
- Retrieving viewing history.
- Removing history records.

15. REVIEWS AND RATINGS TESTING

Verify:

- Submitting ratings.
- Submitting reviews.
- Editing permitted reviews.
- Deleting personal reviews.
- Preventing unauthorized review changes.
- Calculating average ratings.
- Moderating reported reviews.

16. ADMINISTRATION TESTING

The administrator interface must undergo dedicated testing.

Test:

- Administrator login.
- Dashboard statistics.
- User management.
- Movie management.
- Genre management.
- Category management.
- Media uploads.
- Review moderation.
- Reports.
- Platform settings.
- Role permissions.

17. DATABASE TESTING

PostgreSQL testing must verify:

- Table creation.
- Column constraints.
- Primary keys.
- Foreign keys.
- Unique constraints.
- Database relationships.
- Data insertion.
- Data updates.
- Data deletion.
- Transaction consistency.
- Database backup.
- Database restoration.

18. API TESTING

Every API endpoint must be tested.

Testing must verify:

- Correct HTTP methods.
- Correct response status codes.
- Valid response structures.
- Input validation.
- Authentication requirements.
- Authorization requirements.
- Error responses.
- Pagination.
- Rate limiting.

Tools may include Postman and automated API testing frameworks.

19. SECURITY TESTING

Security testing must cover:

- SQL injection prevention.

- Cross-site scripting prevention.
- CSRF protection.
- Password security.
- Session security.
- Unauthorized access.
- Role permissions.
- File upload validation.
- API rate limiting.
- Sensitive information exposure.
- Streaming access authorization.

Security testing must be performed only against authorized FILMORA environments.

20. PERFORMANCE TESTING

Performance testing will measure:

- Homepage loading time.
- Movie search response time.
- API response time.
- Database query performance.
- Concurrent user capacity.
- Video startup time.
- Streaming stability.
- Media delivery performance.

Testing tools may include:

- k6.
- Apache JMeter.
- Lighthouse.

20.1 Performance objectives

Initial proposed targets:

Metric	**Target**

Metric	Target
API response time	Under 500 ms for typical requests
Homepage loading	Under 3 seconds on a suitable connection
Database query time	Under 200 ms for typical queries
Video startup	Under 5 seconds under suitable conditions
Availability	Target 99.5% monthly

These are initial engineering targets, not guaranteed results.

21. RESPONSIVE DESIGN TESTING

Verify the website on:

- Desktop computers.
- Laptops.
- Tablets.
- Mobile phones.

Test:

- Navigation.
- Movie grids.
- Forms.
- Buttons.
- Movie player.
- User dashboard.
- Admin dashboard.

22. BROWSER COMPATIBILITY TESTING

FILMORA should support modern versions of:

- Google Chrome.
- Mozilla Firefox.
- Microsoft Edge.
- Safari.

Video playback must also be tested against browser-specific media format support.

23. ACCESSIBILITY TESTING

Verify:

- Keyboard navigation.
- Screen reader compatibility.
- Color contrast.
- Form labels.
- Image alternative text.
- Focus indicators.
- Video captions.
- Reduced-motion support.

24. ERROR HANDLING TESTING

Test how the platform responds to:

- Network failures.
- Database connection failures.
- Invalid requests.
- Missing movies.
- Expired sessions.
- Failed uploads.
- Unsupported video formats.
- Streaming interruptions.
- Server errors.

Users must receive understandable messages without exposure of internal technical information.

25. USER ACCEPTANCE TESTING

User Acceptance Testing (UAT) will determine whether FILMORA meets its intended requirements.

Test participants will evaluate:

- Account creation.
- Login experience.
- Website navigation.
- Movie discovery.

- Video playback.
- User dashboard.
- Watchlist.
- General usability.

Feedback must be documented and reviewed before release.

26. BUG TRACKING AND REPORTING

Every identified defect must be recorded.

Required information:

- Bug identification number.
- Bug title.
- Description.
- Steps to reproduce.
- Expected result.
- Actual result.
- Severity.
- Priority.
- Assigned developer.
- Current status.

26.1 Bug severity

Severity	Description
Critical	Major system failure or serious security issue
High	Important functionality unavailable
Medium	Feature works incorrectly
Low	Minor visual or usability issue

27. TESTING DOCUMENTATION

The project must maintain:

- Test plans.
- Test cases.

- Test results.
- Bug reports.
- Security reports.
- Performance reports.
- Acceptance reports.
- Release approval records.

28. PRODUCTION READINESS CHECKLIST

Before launching FILMORA:

- Registration works.
- Login works.
- Logout works.
- Password recovery works.
- User permissions are correct.
- Movie management works.
- Search works.
- Movie details work.
- Full-length playback works.
- Streaming authorization works.
- Watchlist works.
- Favorites work.
- Watch history works.
- Admin dashboard works.
- Database backups work.
- Security tests pass.
- Performance tests pass.
- Responsive design tests pass.
- Privacy requirements are satisfied.
- Content distribution rights are verified.
- Critical and high-priority bugs are resolved.

29. QUALITY ASSURANCE DEVELOPMENT PHASES

Phase	Activity
1	Prepare testing environment
2	Prepare test cases
3	Perform unit testing
4	Perform integration testing
5	Perform authentication testing
6	Perform movie management testing
7	Perform streaming testing
8	Perform database testing
9	Perform API testing
10	Perform security testing
11	Perform performance testing
12	Perform responsive testing
13	Perform user acceptance testing
14	Resolve identified defects
15	Perform final regression testing
16	Approve production release

FINAL QUALITY ASSURANCE PRINCIPLE

FILMORA must not be released until its essential functionality, security controls, database operations and video streaming capabilities have been tested.

The testing process must continue after launch through monitoring, user feedback and regular regression testing.

DOCUMENT STATUS: PROPOSED — AWAITING APPROVAL

FILMORA — YOUR WORLD OF MOVIES.